Name: _____ Date: _____ Period: _____

Unit 2 Park Tour: Matter and Energy in Our National Parks

Learning Intention: Today we will be learning how the ideas from Unit 2 show up in the national parks we studied so we can explain how energy and matter shape the places worth protecting.

Success Criteria:

- I can identify evidence of heat transfer, phase changes, and rock formation in a video clip.
- I can explain how each Day 1 rock formed using its rock type.
- I can use evidence to explain why a park is worth protecting.

Word Bank: conduction • convection • density • phase change • magma • lava • crystals • igneous • sedimentary • metamorphic • weathering • erosion • deposition • uplift • rock cycle

Part A: Evidence Tracker

You will watch short clips from five parks we studied this unit. During each clip, write what you **observe**. When the video pauses, answer the question using at least one word from the word bank.

Clip and Question	What I Observe	My Answer
1. Yellowstone Where does the energy that heats the water come from? What phase change causes a geyser to erupt?		
2. Hawai‘i Volcanoes What phase change happens as lava cools? What type of rock does it form?		
3. Yosemite Granite is igneous, like Hawai‘i's lava rock, but it formed in a different place. Where did it cool, and how do its crystals show that?		
4. Badlands What type of rock makes up the Badlands layers? What process is wearing them away?		
5. Grand Canyon The Grand Canyon formed in four steps: Deposition, Uplift, Downcutting, Erosion. For each step, describe one piece of evidence or one event from the video.		

Part B: Back to Day 1

On Day 1, you observed rocks from four parks without naming them. Now name each rock type and explain how it formed.

Day 1 Site	Rock Type (ign, sed, or meta)	How It Formed
Site A: Yosemite granite		
Site B: Hawai‘i Volcanoes basalt and obsidian		
Site C: Badlands sandstone and shale		
Site D: Great Smoky Mountains slate or phyllite		

Part C: What Makes a Place Worth Protecting?

Choose one park from today. Use the science you learned in this unit to explain why it is worth protecting.

Claim: _____ is worth protecting because _____

_____

Evidence (something you saw in the video):

_____

_____

Reasoning (connect your evidence to a Unit 2 idea):

_____

_____

_____